

CHADRON, NE, USA

WMO: 725636

Lat: 42.837N

Lon: 103.098W

Elev: 1004

StdP: 89.83

Time Zone: -7.00 (NAM)

Period: 00-19

WBAN: 24017

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-22.3	-18.8	-25.8	0.4	-22.1	-22.3	0.6	-18.5	14.0	-2.4	12.3	-0.5	2.2	250	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	37.4	19.0	35.4	18.6	33.5	18.4	21.6	31.0	20.7	30.4	20.0	29.7	5.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	15.5	25.3	17.8	14.5	24.2	17.0	13.7	23.9	67.8	30.6	64.4	30.4	61.5	29.6	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
12.7	11.1	9.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-28.8	40.3	2.5	1.9	-30.6	41.7	-32.1	42.8	-33.5	43.9	-35.3	45.3		
			-29.0	23.3	2.5	1.2	-30.8	24.1	-32.2	24.9	-33.6	25.6	-35.4	26.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-3.5	-3.2	3.3	8.0	13.0	19.6	24.4	22.8	17.3	9.1	2.4	-4.1
	DBStd	11.31	6.35	7.07	6.41	5.30	4.98	4.28	3.43	3.71	4.88	5.34	6.01	6.67
	HDD10.0	1884	417	371	219	96	27	0	0	0	4	79	233	436
	HDD18.3	3812	676	604	466	309	176	34	2	8	76	288	478	694
	CDD10.0	1579	0	1	12	37	120	289	445	397	223	52	4	0
	CDD18.3	467	0	0	0	1	11	73	189	147	46	2	0	0
	CDH23.3	6874	0	0	16	69	241	1090	2559	1959	836	99	3	0
	CDH26.7	3453	0	0	1	11	78	501	1424	1029	384	26	0	0
(14) Wind	WSAvg	4.0	3.9	3.9	4.2	4.7	4.4	4.0	3.8	3.8	3.9	3.9	3.8	3.7
(15) Precipitation	PrecAvg	435	8	12	27	54	78	75	53	36	38	35	14	10
	PrecMax	591	21	42	65	101	191	165	122	105	121	82	29	28
	PrecMin	257	2	0	4	11	26	11	7	1	7	2	2	0
	PrecStd	82	6	10	17	24	44	38	25	23	26	22	7	6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.3	18.6	25.9	28.0	32.2	38.0	40.5	38.6	36.2	29.8	23.3	15.6
		MCWB	6.1	8.7	11.2	12.9	17.3	18.7	19.7	17.7	17.9	13.8	10.5	6.3
	2%	DB	11.4	13.9	21.9	25.5	28.7	34.6	38.0	36.6	33.5	26.3	19.1	11.7
		MCWB	4.8	5.9	9.9	12.2	16.4	18.4	19.4	18.7	16.7	12.9	8.7	4.9
	5%	DB	8.0	10.2	18.3	22.2	26.0	32.0	36.1	34.6	31.3	22.8	16.1	8.1
		MCWB	2.9	4.0	8.6	11.4	15.3	18.7	19.1	18.4	16.1	11.8	7.3	3.0
	10%	DB	5.7	7.1	14.6	18.7	23.1	29.4	33.8	32.6	28.4	19.6	12.7	5.3
		MCWB	1.8	2.2	6.6	9.9	14.2	18.1	19.0	18.4	15.5	10.6	5.7	1.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.8	8.8	12.3	14.7	19.7	22.9	22.8	21.9	20.5	15.5	11.0	7.4
		MCDB	13.5	18.6	23.0	24.6	28.2	30.3	32.4	31.7	30.5	25.8	21.9	14.9
	2%	WB	4.9	6.1	10.2	13.1	17.6	21.2	21.8	20.9	18.6	13.9	9.1	5.0
		MCDB	11.0	13.3	20.6	23.3	25.9	29.3	31.4	31.0	28.0	23.5	18.3	11.1
	5%	WB	3.3	4.3	8.8	11.6	16.1	19.9	21.0	20.1	17.5	12.4	7.6	3.2
		MCDB	7.7	9.5	17.7	21.3	23.8	28.7	30.8	29.9	28.0	21.0	15.8	8.0
	10%	WB	1.9	2.5	7.1	10.3	14.8	18.8	20.3	19.2	16.5	11.1	6.0	1.5
		MCDB	5.4	6.7	14.2	18.4	22.2	27.7	30.2	28.7	26.3	18.7	12.1	5.1
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	13.8	15.2	14.7	14.6	15.9	17.0	17.1	17.5	16.0	15.5	13.4
		MCDBR	16.6	18.0	21.8	21.4	19.6	20.0	20.2	19.9	21.7	22.2	21.1	17.1
		MCWBR	10.9	10.8	11.6	10.4	9.1	7.8	6.3	6.3	8.2	10.8	11.9	11.3
	5% WB	MCDBR	15.6	17.0	20.4	19.8	17.0	17.0	17.0	16.9	18.2	19.6	20.3	16.7
		MCWBR	10.7	10.7	11.2	10.0	8.5	8.1	6.5	6.2	7.3	9.8	11.6	11.3
(40) Clear-Sky Solar Irradiance	taub		0.256	0.264	0.285	0.314	0.336	0.349	0.372	0.369	0.327	0.293	0.268	0.255
	taud		2.572	2.547	2.526	2.440	2.448	2.431	2.401	2.390	2.490	2.560	2.552	2.557
	Ebn at Noon		913	957	968	955	936	920	894	887	906	907	886	880
	Edh at Noon		65	79	91	108	110	113	115	113	94	77	66	60
(44) All-Sky Solar Radiation	RadAvg		2.09	2.99	4.18	5.18	6.11	7.19	7.16	6.19	4.96	3.34	2.31	1.75
	RadStd		0.11	0.17	0.32	0.30	0.51	0.47	0.25	0.23	0.28	0.30	0.14	0.12

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page